Percolation and pattern formation in collective migration in Gliomas

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ABSTRACT

I. MODEL

We consider a system composed of two cell phenotypes: round (passive) and elongated (motile). The latter can exhibit variable aspect ratios, constrained within the range ϕ, with $1 < \phi < \phi_{max} = 2.7$. A novel aspect of our model is the plasticity of cell phenotype: round cells can elongate, and elongated cells can contract. Furthermore, the intrinsic speed of a cell increases linearly with the aspect ratio, vanishing for round cells and reaching unity for those with maximum aspect ratio.

The basic idea of elongation is the tendency of cells to deform and migrate when local space is available. On the other hand, elongated cells shrink if they experience enough force. To implement these ideas to our model, we apply the following mechanisms:

- A round cell attempts to elongate by sampling a random angle $\theta \in [0, 2\pi)$. For this direction, we calculate the overlap of the potentially deformed cell with its neighbors. If the overlap is below a defined threshold for all neighbors, the angle becomes a candidate. We repeat this procedure for 10 trials. Finally, we select the candidate direction that minimizes the total overlap. If no candidate satisfies the threshold condition, the cell remains round.

- Elongated cells exhibit two behaviors. They can shrink if the force exerted by neighbors is sufficient to arrest motion in the direction of the intrinsic velocity. If this does not happen, they can try to continue their elongation in the direction of motion.

For the initial conditions, the system starts with all cells in the round state, placed randomly. We wait some time for the system to relax and avoid large overlaps before the elongation dynamics begin.

II. CLASSIFICATION OF STATES

Figure 1 shows examples of the steady state for different system densities. At low densities, the system is dominated by elongated cells, which can be either isolated or form small clusters. As the density increases, the fraction of elongated cells decreases, and they are more likely to form clusters rather than move in isolation, while the population of round cells grows, ultimately reaching a state where the entire system is composed of round cells.

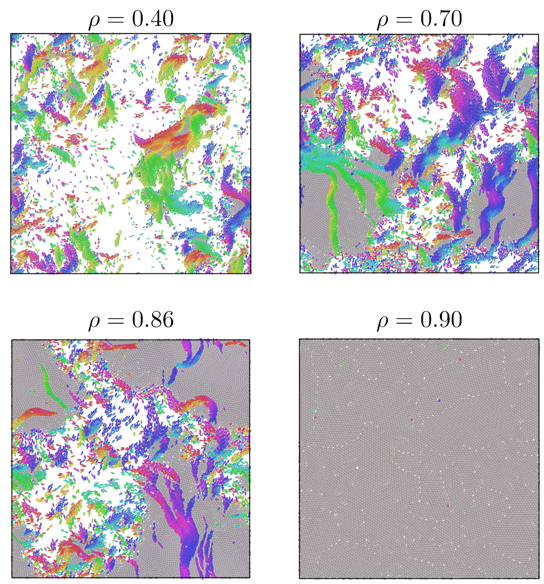


FIG. 1: Examples of the steady state for different system densities.

Broadly speaking, two different states can be identified. The first one, where cells find space to elongate (as at low densities in Fig. 1), which we associate with an unjammed state, and a second one where all cells remain round, forming a jammed state (as in $\rho = 0.9$ in Fig. 1).

In order to classify each realization as jammed or unjammed, both the largest cluster size of round cells (S_{max}) and the fraction of active cells (f_a) are analyzed. It is expected that unjammed states have a significant fraction of elongated cells, while jammed states are characterized by a giant cluster of round cells. Fig. 2 (a) shows a scatter plot of S_{max}/N vs f_a, and (b) the distribution of f_a. We perform a DBSCAN clustering on the data in (a), choosing an ϵ that separates the data into two groups without noise. This choice is consistent with the valley observed in (b), and we use this criterion to classify the states, as shown in Fig. 2 (c). From this classification, we compute the jamming probability as the fraction of realizations classified as jammed for each density, shown in Fig. 2 (d). A clear transition between unjammed and jammed states can be observed as the density increases: the system evolves from predominantly unjammed configurations at low densities to fully jammed states at high densities. The relatively sharp change in the jamming probability indicates a well-defined transition between these two regimes.

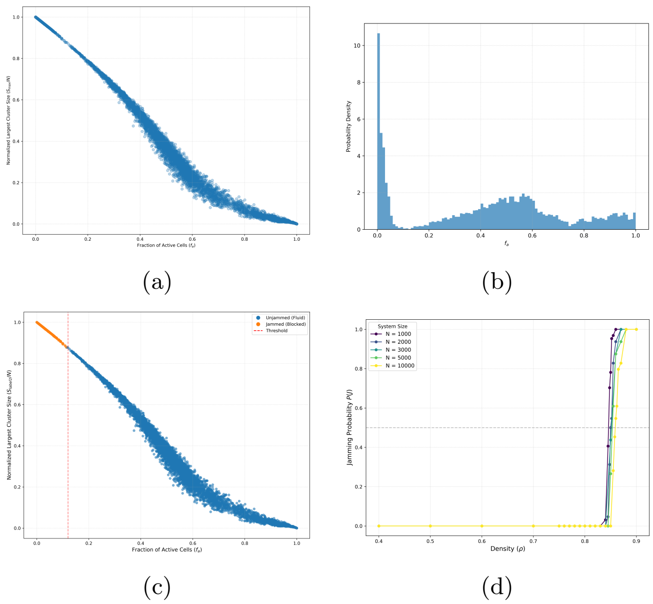


FIG. 2: Classification of every realization (for all N and ρ), using the largest cluster of round cells (S_{max}) and the fraction of active cells (f_a). (a) Scatter plot of S_{max}/N vs f_a. (b) Global distribution of f_a. (c) Classification of states. (d) Jamming probability as a function of density.

III. STRUCTURE OF CLUSTERS

To characterize the spatial organization of the system, we analyze the cluster structure formed by each type of cells. In order to do so, we define that two cells belong to the same cluster if they fulfill two conditions: (i) They are of the same type (both elongated or both round), and (ii) they are close enough to interact, allowing for a small tolerance δ, such that cells that are within a distance slightly larger than the interaction threshold are still considered connected.

Figure 3 presents the evolution of several cluster observables as a function of density. Panel (a) shows the normalized size of the largest cluster (S_1). At low densities, elongated cells form extended structures, with the largest cluster reaching approximately half of the system size, and exhibiting a maximum around $\rho \approx 0.6$. However, S_1 decreases at higher densities, eventually vanishing as the system becomes jammed. In contrast, round cells exhibit a clear percolation transition: the giant component grows continuously at first, followed by an abrupt jump to unity at high densities.

Another important feature to measure is the second largest cluster. Panel (b) displays the normalized size of the second largest cluster (S_2), while panel (c) shows the ratio $\frac{S_2}{S_1}$. For round cells, S_2 exhibits a peak reaching approximately 6% of the system size just before the transition. This indicates that, slightly below the percolation threshold, the system is characterized by the competition of multiple macroscopic clusters. For elongated cells, the ratio remains relatively large, $S_2/S_1 \approx 0.3$ at $\rho = 0.4$, indicating a fragmented structure without a dominant component. As density increases, this ratio decreases and reaches a minimum in the range $\rho \in [0.6, 0.76]$, consistent with the maximum of S_1 and the minimum of S_2.

Finally, panel (d) shows the mean number of clusters. At low densities, elongated cells form a large number of small clusters, reflecting the presence of many isolated or weakly connected cells. As density increases, the number of clusters decreases while S_1 increases, indicating the progressive aggregation into larger structures. Near the transition, the number of clusters drops sharply. In contrast, round cells maintain a relatively constant number of clusters, between 100 and 200 (corresponding to 1%–2% of the system size), before collapsing to zero at the transition point.

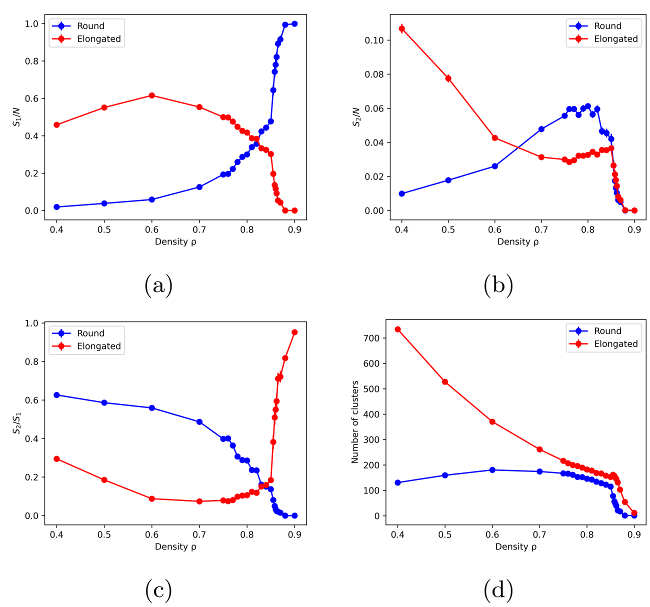


FIG. 3: Cluster metrics as a function of density for both clusters of elongated and round cells. (a) Largest cluster size (S_1). (b) Second largest cluster size (S_2). (c) Ratio $\frac{S_2}{S_1}$. (d) Mean number of clusters.

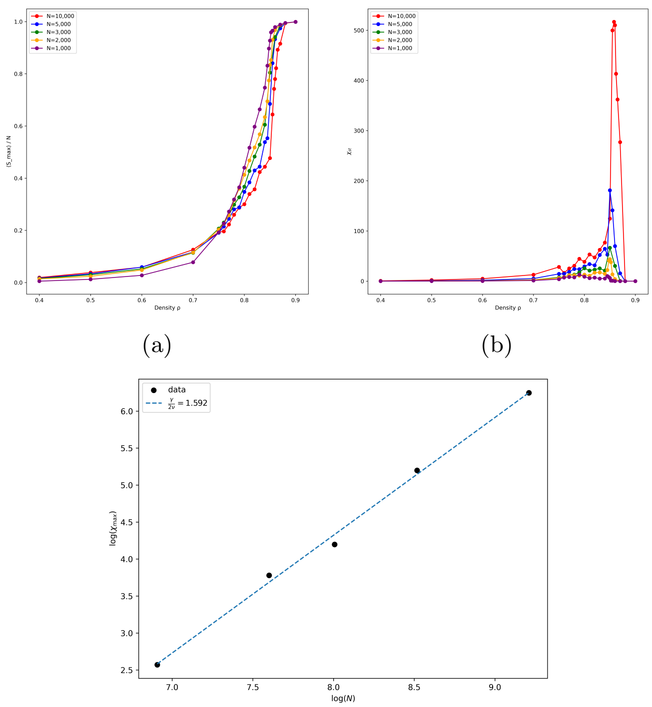
IV. PERCOLATION OF ROUND CELLS

A. Size of largest cluster and Susceptibility

As mentioned above, round cells exhibit a percolation transition. We can see how the largest cluster size vary with density for different values of N in Fig. 4 (a). To gain insight into the nature of this transition, we analyze the susceptibility χ, defined as the fluctuations of the normalized size of the largest cluster S_1:

$$\chi = N(\langle S_1^2\rangle - \langle S_1\rangle^2) \tag{1}$$

Fig. 4 (b) shows the susceptibility for different system sizes N. It can be observed that for every size, χ exhibits a pronounced peak near the critical density $\rho \approx 0.86$.



(a) (b)

(c)

FIG. 4: Statistics of percolation for round cells for different N over ρ. (a) Largest cluster size (S_{max}/N). (b) Susceptibility (χ). (c) Finite-Size Scaling of the susceptibility peaks.

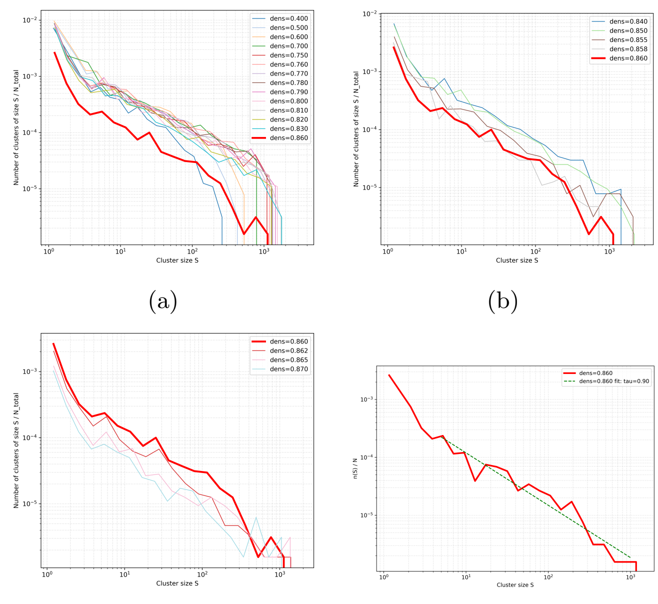
The scaling of the susceptibility peak maximum provides crucial information about the universality class of the transition. For a finite system of size L, the peak of χ near the critical point behaves as $\chi^{max} \propto L^{\gamma/\nu}$. In our case, we have a box of size L in 2D. Therefore, $N \propto L^2$, or $L \propto N^{1/2}$, and then $\chi_{st}^{max} \propto N^{\gamma/2\nu}$. Figure 4 (c) displays the scaling of the susceptibility peaks with N in a log-log plot. A linear fit yields an effective exponent of $\frac{\gamma}{2\nu} = 1.592$.

For standard percolation in 2D, the exponent is $\frac{\gamma}{2\nu} \approx 0.896$. This is very different from the value we obtain. Moreover, for a standard discontinuous transition (first-order), the susceptibility is expected to scale linearly with the system volume ($\chi \propto L^2 \propto N$). Our observed exponent of ≈ 1.592 is super-extensive (greater than 1). This implies that the fluctuations grow faster than the system size itself, indicating an anomalously violent regime.

B. Cluster Size distribution excluding the giant component

Finally, we analyze the cluster size distribution n_s (excluding the giant component) in the steady state. According to percolation theory, this distribution is expected to exhibit a power-law behavior $n_s(s) \propto s^{-\tau}$ near the critical point ρ_c.

Figure 5 displays these distributions for different densities, with the critical density ρ_c (derived from the susceptibility peak) highlighted in every panel.



(a) (b)

(c) (d)

FIG. 5: Cluster size distribution without the giant component. (a) and (b) correspond to densities below ρ_c: (a) for densitiesn significantly below and (b) approaching ρ_c. Panel (c) shows the regime of high densities. Panel (d) displays the fit of the power law of the cluster size distribution in the critical density.

At low densities (a), the distribution follows a power law for small sizes but is truncated by a cut-off at larger sizes. As the density approaches the critical point (b), this cut-off tends to disappear. At ρ_c, the cut-off effectively disappears, and the distribution exhibits a pure power-law decay.

The physical interpretation may be that below the critical point, the correlation length is finite, so the maximum cluster size is limited. In addition, at low densities, the curve lies slightly above the critical one. This happens because there is a large number of small clusters in the system.

At the critical point, fluctuations of all sizes exist. However, once the threshold is crossed (panel c), the giant component absorbs most of the cells. Therefore, the distribution of the remaining clusters decays rapidly, leaving only small fragments isolated from the giant structure.

If we fit the power-law in the critical density, we obtain $\tau \approx 0.90$ for $N = 10.000$, as it is shown in figure 5 (d). However, this exponent change with N due to finite-size effects.

In usual percolation, the relation $2 < \tau < 3$ is typically be satisfied. Therefore, our result is in contrast with usual percolation. This may be associated with the explosive nature of the transition. There are a lot of macroscopic clusters that grow independently, creating an excess of large clusters just before the abrupt formation of the giant component.

C. Cluster Size distribution with the giant component

D. Different Mechanism for Percolation

These results show that, although round cells percolate, the mechanism is not the usual one. The combination of an abrupt jump in the largest cluster size and the power law in the distributions is a key feature of hybrid phase transitions. In percolation, we have found two different type of percolations that can be associated with our problem.

On one hand, explosive percolation happens when the system avoids the creation of a giant cluster by minimizng the size of the clusters that are being combined. One example is the product rule: two links are chosen and only the one with the minimum result between the product of each cluster is kept. This can be associated with the appearance of a huge second cluster size before the transition.

On the other hand, bootstrap percolation has an analogy with the mechanism that our system follows. In bootstrap percolation, nodes are either active or inactive, and they deactivate if they have less than k active neighbors. The analogy arises if we think of active nodes as round cells. Thus, cells deactivate (elongate) if they have few active neighbors (if they have enough space).

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V. ANALYSIS OF THE UNJAMMED BRANCH

As previously established, the system can reach two distinct steady states: a jammed state, where all cells remain round and the system is structurally blocked, and an unjammed state, characterized by cell elongation and collective motion. In this section, we provide an analysis of the unjammed branch. While Section III examined the cluster structure across all realizations, averaging near the transition can lead to an artificial suppression of the physical observables. For instance, the system polarization vanishes at high densities primarily due to the increasing prevalence of jammed configurations. Therefore, focusing exclusively on the active branch allows us to characterize the intrinsic properties of the fluid phase.

A. Analysis of clusters of elongated cells

The behavior of the elongated phase as a function of density ρ is summarized in Figure 6. Panel (a) shows that as the density increases, the total number of clusters normalized by the system size decreases. This suggests that the system tends to coalesce into fewer structures as the available space becomes more constrained. In panel (b) it can be seen that the size of the largest cluster is consistently much larger than the second largest one across most of the density range. Interestingly, S_{max} reaches a maximum value near $\rho = 0.6$. Outside this range, for both very low and very high densities, the size of the dominant cluster decreases. Finally, panel (c) displays the ratio $\frac{S_2}{S_{max}}$, while panel (d) the parameter $\frac{S_{max}-S_2}{S_{max}+S_2}$. In both, we can see that he region between $\rho = 0.6$ and $\rho = 0.77$ marks the point of maximum dominance of the largest cluster, indicating the formation of a single, well-defined oncostream.

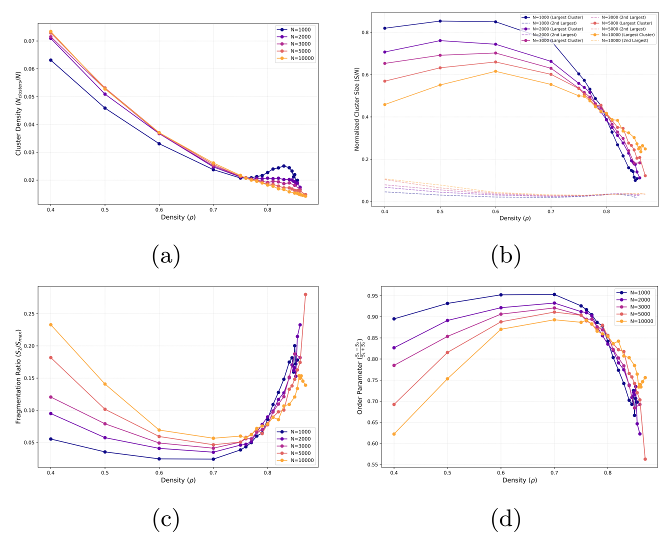


FIG. 6: Analysis of the elongated clusters in the unjammed branch for $N = 1000, 2000, 3000, 5000, 10000$ as a function of density. (a) Cluster density. Number of total elongated clusters over number of total cells ($\frac{N_{clusters}}{N}$). (b) Both largest (S_{max}) and second largest (S_2) clusters. (c) Fragmentation ratio (S_2/S_{max}). (d) Order Parameter ($\frac{S_{max}-S_2}{S_{max}+S_2}$).

Based on these observations, we can identify distinct regimes for the active phase. At low densities, the system forms many independent clusters and the largest one does not yet dominate the system's dynamics. This makes that this regime exhibits a mosaic-like structure. Approximately between $\rho = 0.6$ and $\rho = 0.7$, appears a giant cluster that governs the collective motion. As the system approaches the jamming point, the number of clusters decreases, but they all become comparable in size.

B. Analysis of polarization of elongated cells

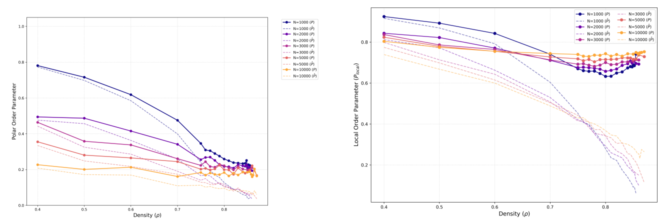
Finally, we can study how ordered the final structures are. In order to do so, we use the well-known polar order parameter (P) calculated using the number of elongated cells as follows:

$$P = \frac{1}{N_e}\sqrt{\left(\sum_{i=1}^{N_e}\sin(\varphi_i)\right)^2 + \left(\sum_{i=1}^{N_e}\cos(\varphi_i)\right)^2},$$

as well as the same polar parameter as before but dividing by all the cells (including the rounds), ($\hat{P}$):

$$\hat{P} = \frac{1}{N}\sqrt{\left(\sum_{i=1}^{N_e}\sin(\varphi_i)\right)^2 + \left(\sum_{i=1}^{N_e}\cos(\varphi_i)\right)^2}.$$

Figure ... displays how this order parameter varies in function of the density for all the values of N. Panel (a) shows the global order parameter while panel (b) the local. For the last, ... EXPLICAR BIEN ESTO.



(a) (b)

FIG. 7: Polarization of the clusters of elongated cells in the unjammed branch for $N = 1000, 2000, 3000, 5000, 10000$ as a function of density. (a) Global Polar Order. (b) Local Polar Order.